

Class #23 - The Dimension of a Vector Space

1. Def: A one-to-one linear transformation from a vector space V onto a vector space W is called an *isomorphism from V onto W* .

iso: _____, morph: _____

2. Example: Let v_1 and v_2 be independent vectors in $\mathbf{R}^3$.

Let $H := \text{Span} \{v_1, v_2\}$. Then, _____ is an isomorphism from _____ onto _____. Hence, H is isomorphic to _____.

3. Example: Verify that the polynomials $1 + 2t^2$, $4 + t + 5t^2$, and $3 + 2t$ are linearly dependent in $\mathbf{P}_2$.

4. Fill in the blank: A vector space with a basis B containing n vectors is isomorphic to _____.

Question: Is it possible for a vector space to have two bases with different number of vectors?

5. Theorem: If a vector space V has a basis $B = \{b_1, b_2, \dots, b_n\}$, then any set in V containing more than n vectors must be _____.

Proof:

6. Theorem: If a vector space V has a basis of n vectors, then every basis of V must consist of exactly _____ vectors.

Proof:

7. Def: If V is spanned by a finite set, then V is said to be *finite-dimensional*, and the *dimension of V* , written as _____, is the number of vectors in a _____ for V .

The dimension of the zero vector space $\{0\}$ is defined to be _____.

If V is not spanned by a finite set, then V is said to be _____-dimensional.

8. Example: $\dim \mathbf{R}^n =$ _____, $\dim \mathbf{P}_n =$ _____, $\dim \mathbf{P} =$ _____.

9. Example: Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbf{R} \right\}$$

10. Example: Find k -dimensional subspaces of $\mathbf{R}^3$ for each $k = 0, 1, 2, 3$.

11. Theorem: Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded to a _____ for H . Also, H is _____-dimensional and

$$\dim H \text{ _____ } \dim V.$$

12. Fill in the blanks: Let V be a n -dimensional vector space with $n \geq 1$. Any linearly independent set of exactly n vectors in V is automatically a _____ for V .

Any set of exactly n vectors that spans V is automatically a _____ for V .

13. Fill in the blanks: Let $A \in \mathbf{R}^{m \times n}$.

$$\text{nullity } A := \dim \text{Nul } A = \text{_____},$$

$$\text{rank } A := \dim \text{Col } A = \text{_____}.$$

Hence,

$$\text{rank } A + \text{nullity } A = \text{_____}.$$